

VISVESVARAYATECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi – 590018



Project Report

On

“CRACK DETECTION AND WATER LEVEL INDICATOR IN DAM”

Submitted in partial fulfilment of the requirement for the award of degree of

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION

Submitted by

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**Approved by AICTE, Permanently Affiliated to VTU, Belagavi, Accredited by NAAC & NBA
2025-2026**

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CERTIFICATE

This is to certify that the project work entitled “**CRACK DETECTION AND WATER LEVEL INDICATOR IN DAM**” is a bonafide work carried out by **NAVEEN B(1AM22EC072), PALLAVI S (1AM22EC089), PRADEEP S N(1AM22EC086), PRATHIBHA H(1AM22EC089)** In partial fulfillment for the award of Bachelor of Engineering in Electronics and Communication of the Visvesvaraya Technological University, Belagavi during the year 2025-26. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

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DECLARATION

We, the students of VII semester B.E in Electronics and Communication Engineering, AMC Engineering College, Bengaluru, hereby declare that the project work entitled “**Crack Detection and Water Level Indicator in Dam**” submitted to the Visvesvaraya Technological University during the academic year 2025-26, is a record of an original work done by us under the guidance of **Dr. R Aruna, Associate Professor** Department of Electronics and Communication Engineering, AMC Engineering college, Bengaluru. This project work is submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Electronics and Communication Engineering. The results embodied in this report have not been submitted to any other university or institute for the award of any degree.

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ACKNOWLEDGEMENT

We would like to thank our chairman **Dr. K. R. Paramahamsa** AMC Engineering College, Bangalore, for providing the necessary infrastructure.

We would like to thank our Principal, **Dr. Yuvaraju B N**, AMC Engineering College, Bangalore, for his kind cooperation.

We sincerely thank **Dr. Jenitta J**, HOD, Dept. of Electronics and Communication Engineering, AMC Engineering College, Bangalore, from the bottom of our heart for his support and understanding.

We consider it privilege to express gratitude to our guide, **Dr. R Aruna**, Associate professor, Dept. Of Electronics and Communication Engineering, AMC Engineering College, Bangalore, for their constant support and valuable suggestions during the course of the Final Project.

We would also like to extend my gratitude to the faculty coordinators of Dept. Of Electronics and Communication Engineering, AMC Engineering College, Bangalore, for their valuable input and encouragement. Their expertise and knowledge have greatly enriched my understanding of the subject matter.

Our deepest gratitude goes to our teachers, who showed infinite patience and understanding till the completion of our Project.

Last, but not the least, our sincere credit to our parents, our friends and to one and all who have directly or indirectly helped us in the successful completion of the Project.

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ABSTRACT

This project presents an integrated system for crack detection and water level monitoring in dams, aimed at improving structural safety and operational efficiency. Dams are critical infrastructure that require constant monitoring to prevent catastrophic failures due to structural weaknesses or improper water level management. The proposed system utilizes Internet of Things (IoT) and sensor-based technology to detect surface cracks and monitor water levels in real-time. Crack detection is achieved using ultrasonic or vibration sensors, which identify structural deformations and transmit data to a central system for analysis. Simultaneously, ultrasonic or float-based water level sensors measure the height of water within the dam reservoir and notify operators when levels exceed or drop below predefined thresholds. This information is continuously relayed via a microcontroller (such as Arduino or Raspberry Pi) to a cloud-based or local display and alert system, enabling timely interventions. By combining both safety features into a single automated system, the project enhances dam management, reduces the risk of human error, and enables predictive maintenance. This smart monitoring approach is scalable, cost-effective, and vital for modern dam safety management, particularly in regions prone to heavy rainfall or seismic activity.

CHAPTER- 1

INTRODUCTION

Dams are monumental structures that serve as lifelines for irrigation, hydroelectric power generation, flood control, and water supply. As critical components of a country's infrastructure, their structural integrity and operational reliability must be ensured at all times. However, due to aging, environmental stress, natural disasters, or continuous pressure from water and sediments, dams are susceptible to structural deterioration, especially crack formation. Simultaneously, the water level within the dam must be constantly monitored to prevent overflows or excessive depletion, which could compromise the dam's functionality or even cause catastrophic failure. In this context, the proposed project—Crack Detection and Water Level Indication System in Dams—aims to provide a smart, reliable, and cost-effective monitoring solution using modern embedded and sensor-based technologies.

Traditionally, dam inspection and monitoring have relied heavily on manual methods and periodic checks. While visual inspection is a common approach, it is limited by accessibility challenges, human error, and infrequent observation intervals. Many small cracks or early signs of damage may go unnoticed until they develop into significant structural risks. Similarly, water level monitoring often involves outdated float-type gauges or manual reading systems that may not provide real-time alerts or automated responses. In the modern age of digital automation and IoT, there is a growing need for smart monitoring systems that provide continuous, real-time feedback and early warnings, allowing for prompt action and disaster prevention.

This project addresses these issues by integrating two essential safety features—crack detection and water level indication—into one seamless system. The crack detection module uses sensors such as ultrasonic transceivers or vibration sensors to monitor changes in structural integrity. These sensors can detect surface-level or internal cracks by measuring alterations in wave patterns or vibrations passing through dam walls. In some advanced setups, strain gauges or fiber optic sensors can also be employed to detect micro-cracks or structural strain. These sensors are connected to a microcontroller, such as an ESP8266 or Raspberry Pi, which collects data and sends it for processing and analysis. Once a crack is detected, an alert can be triggered to the central control room, notifying dam authorities to inspect and initiate timely repairs.

Parallelly, the water level monitoring module works using ultrasonic distance sensors or float switches that measure the height of water in the reservoir. These sensors provide continuous

real-time data on the water level and are also connected to the microcontroller. If the water level rises above or falls below safe thresholds, the system sends immediate alerts via display screens, mobile notifications, or alarm systems. This helps in managing dam gate operations, warning downstream communities, and preventing potential flooding or drought scenarios.

The key advantage of this dual-function system lies in its automation, scalability, and cost-effectiveness. It minimizes human involvement while ensuring 24/7 surveillance, making it especially useful for remote or high-risk dams. It is also modular, allowing for customization based on dam size, risk levels, and environmental conditions. Additionally, the data collected can be logged and analyzed over time to predict structural health trends, optimize maintenance schedules, and support decision-making processes.

1.1 HISTORY OF FIELD OF INTEREST

The monitoring and maintenance of dams have always been of critical importance due to their role in water storage, flood control, irrigation, and power generation. Over time, advancements in engineering and technology have significantly shaped the methods used for crack detection and water level monitoring.

1. Early Methods and Traditional Monitoring

In the early 20th century, dam inspections were primarily based on visual observation and manual measurements. Engineers would conduct periodic site visits to examine surface cracks, leakage, seepage patterns, and the overall physical condition of the structure. Water level monitoring was typically done using staff gauges, simple float systems, or mechanical devices that required human reading and interpretation. While these methods were the only available options at the time, they had several limitations—primarily low accuracy, irregular data collection, and reliance on human judgment.

2. The Emergence of Instrumentation(1950s–1980s)

With the advancement of civil engineering practices and increasing dam construction during the post-war industrial boom, the need for more reliable monitoring techniques became apparent. In the 1950s to 1980s, engineers began installing basic instrumentation systems in dams, such as:

- Strain gauges to detect stress and deformation in dam materials.
- Vibrating wire piezometers to measure pore pressure.

- Mechanical extensometers to measure crack widths and movements.

During this time, hydraulic and pneumatic water level sensors started replacing older manual methods, offering more accurate and automated measurements. However, data was still mostly collected manually and analyzed periodically.

3. Digital Revolution and Early Automation (1990s–2000s)

The rise of digital electronics and computing in the 1990s led to a major transformation in dam monitoring systems. Crack detection evolved with the use of:

- Ultrasonic sensors which could send high-frequency waves through concrete and detect internal voids or cracks.
- Infrared thermography allowing engineers to visualize temperature variations caused by structural weaknesses or moisture intrusion.
- Laser scanning and photogrammetry enabling surface mapping and detection of geometric changes in dam structures.

Water level monitoring systems also became more advanced with the introduction of pressure transducers, float sensors, and radar-based level sensors that offered real-time data logging and remote access capabilities. These innovations marked the beginning of automated dam monitoring systems, although they were still expensive and mostly used in large-scale or high- risk projects.

4. Modern Era: IoT, Wireless Systems, and Smart Monitoring (2010–Present)

In the past decade, the field has witnessed rapid advancement driven by the Internet of Things (IoT), microcontrollers (like Arduino, Raspberry Pi), and wireless communication technologies. These have made it possible to develop low-cost, real-time, scalable monitoring systems that can detect cracks, record water levels, and transmit alerts instantly to centralized or cloud-based systems.

Today, smart crack detection includes the use of:

- MEMS sensors (Micro-Electro-Mechanical Systems) for detecting fine structural movements.
- Fiber optic sensors that detect strain, pressure, and crack propagation over long distances.
- Machine learning and AI algorithms that predict structural behavior and potential failure risks based on collected data.
- Ultrasonic and radar sensors with real-time distance measurement.

- Solar-powered wireless systems that can operate in remote locations without power supply.
- Integration with mobile apps, SCADA systems, and emergency response networks for immediate alerting and control.

1.2 MOTIVATION OF THE PROJECT

Dams play a vital role in modern society by supporting water storage, irrigation, hydroelectric power generation, and flood control. However, due to constant pressure from stored water, environmental stress, aging infrastructure, and potential natural disasters like earthquakes and heavy rainfall, dams are vulnerable to structural issues such as crack formation and uncontrolled changes in water levels. These issues, if not identified early, can lead to devastating consequences— including dam failure, property damage, environmental destruction, and loss of life. Traditional monitoring methods, which rely heavily on manual inspection and mechanical gauges, often fall short in terms of accuracy, timeliness, and reliability. Cracks may go unnoticed until they expand significantly, and water level changes may not trigger warnings quickly enough to allow preventive action. Moreover, in remote or hard-to-access dam locations, manual monitoring is impractical and inconsistent.

The motivation behind this project is to design and implement a smart, automated system that combines real-time crack detection with continuous water level monitoring using modern sensor and IoT technologies. By utilizing tools like ultrasonic sensors, vibration sensors, float or ultrasonic level sensors, and microcontrollers (such as Arduino or ESP8266), the system can accurately monitor dam health and instantly alert authorities when abnormalities are detected. This integrated approach ensures early detection of structural weaknesses, efficient water level management, and rapid response to potential risks. It reduces the reliance on human labor, improves safety, minimizes maintenance costs, and supports sustainable infrastructure management. Ultimately, the project aims to protect lives, reduce disaster risks, and contribute to the modernization of dam safety practices using accessible and cost-effective technology.

1.3 OBJECTIVE OF THE PROJECT

The primary objective of this project is to design and implement an advanced monitoring system that ensures the safety, structural integrity, and efficient operation of dams by detecting cracks and continuously monitoring water levels. The specific objectives of the project are:

- 1. Early Detection of Cracks:** To detect surface-level and internal cracks in dam structures using ultrasonic, vibration, or other sensor technologies. This enables early identification of potential structural damage, preventing further deterioration and failure.
- 2. Continuous Water Level Monitoring:** To monitor the water levels within the dam reservoir in real-time using advanced sensors (e.g., ultrasonic or float-based sensors). This ensures safe and efficient water management, preventing overflow or inadequate water levels during critical periods.
- 3. Automated Alerts and Notifications:** To provide instant alerts to dam operators or authorities when cracks or unsafe water levels are detected. This allows for immediate corrective action to be taken, reducing the risk of failure or damage.
- 4. Reduction of Manual Inspection:** To minimize the need for frequent manual inspections by implementing an automated system that continuously monitors the dam's condition. This enhances efficiency and reduces human error.
- 5. Cost-Effective and Scalable System:** To develop a cost-effective, modular, and scalable monitoring system that can be easily deployed across various types of dams, from large-scale structures to smaller, less-resourced facilities.
- 6. Data-Driven Decision Making:** To collect and store continuous data for long-term analysis, helping engineers and authorities make informed decisions regarding maintenance schedules, safety measures, and infrastructure improvements.
- 7. Enhanced Safety and Disaster Prevention:** To ensure the safety of the dam and surrounding communities by detecting potential issues before they escalate, contributing to disaster prevention and emergency preparedness.

By achieving these objectives, the project aims to provide a reliable, efficient, and automated solution for monitoring dam safety, ensuring sustainable dam operation, and ultimately protecting both the infrastructure and the communities that rely on it.

1.4 PROBLEM STATEMENT

Structural Integrity Concerns: Dams are exposed to various stress factors such as water pressure, aging, and environmental conditions, which can lead to crack formation. These cracks, if undetected, can compromise the dam's structural integrity, posing a significant risk to both the dam and surrounding areas.

1. Delayed Detection of Cracks: Traditional manual inspection methods often fail to detect cracks in a timely manner, leading to delayed identification of potential structural issues. Early-stage cracks may go unnoticed, allowing them to worsen over time, increasing the risk of catastrophic failure.

2. Integration Gap: Existing monitoring systems often operate in silos, focusing either on crack detection or water level monitoring separately. There is a lack of integration between these systems, preventing a holistic view of the dam's health and water management.

3. Data Analysis: The data collected from manual inspections or outdated systems is often not fully utilized for predictive maintenance or long-term trend analysis, missing opportunities to proactively address structural issues before they escalate.

4. Real-Time Alerts: Current systems often lack real-time alerts and automated notifications when cracks or dangerous water levels are detected, delaying response times and reducing the effectiveness of emergency measures.

This project aims to address these issues by developing a comprehensive, integrated monitoring system that provides real-time crack detection, water level monitoring, predictive analysis, and instant alerts, ensuring continuous, accurate, and timely management of dam safety.

1.5 SUMMARY

This chapter highlights the importance of dam safety and the need for modern monitoring systems to ensure structural integrity and efficient water management. It discusses the limitations of traditional manual inspections and introduces a smart, sensor-based system for real-time crack detection and water level monitoring. The project aims to enhance safety, reduce human error, and enable timely alerts through IoT-based automation. Overall, it emphasizes a cost-effective, integrated solution for continuous dam surveillance and disaster prevention.

CHAPTER- 2

LITERATURE REVIEW

2.1 RELATED WORK

[1]. A Reservoir Dam Monitoring Technology Integrating Improved ABC Algorithm and SVM Algorithm

The paper titled “A Reservoir Dam Monitoring Technology Integrating Improved ABC Algorithm and SVM Algorithm” by Yunqian Xu, Tengfei Bao, Mingdao Yuan, and Shu Zhang (2025) introduces an intelligent and comprehensive approach for the monitoring of reservoir dam structures. The authors address the growing need for reliable, real-time monitoring systems in dam safety management, especially in the face of increasing environmental pressures and aging infrastructure. To achieve this, they propose a hybrid technique that combines an improved Artificial Bee Colony (ABC) algorithm with a Support Vector Machine (SVM) classifier. The improved ABC algorithm is utilized for optimizing parameter selection and enhancing the convergence speed and solution accuracy, while the SVM algorithm is responsible for classifying and analyzing the monitored data to detect anomalies or potential risks. Together, these algorithms form a robust framework capable of processing complex and large-scale sensor data collected from dam environments.

The monitoring system designed in this research takes into account both algorithm-specific and environmental data parameters, allowing for dynamic adaptation and decision-making based on current dam conditions. This integration results in several notable advantages. Most significantly, the system demonstrates high prediction accuracy, ensuring that early warnings can be generated with greater confidence. Additionally, it supports real-time monitoring, enabling operators to respond swiftly to potential threats or abnormalities. The system also exhibits strong data processing capabilities, making it suitable for deployment in diverse and demanding field conditions.

Despite its strengths, the authors acknowledge certain limitations of the proposed approach. One primary challenge lies in the system's dependence on the quality and continuity of input data. Inaccurate or incomplete sensor readings may lead to errors in detection or classification. Furthermore, the model's sensitivity to environmental variations—such as temperature, humidity, or structural vibrations—could affect its consistency and reliability. These factors highlight the importance of deploying high-quality sensors and implementing data preprocessing techniques to enhance system performance.

Overall, the study presents a significant advancement in the field of dam monitoring technologies. By effectively integrating optimization and machine learning algorithms, the proposed method offers a reliable and intelligent solution for enhancing dam safety and management practices. This work lays a strong foundation for future research and practical implementation in real-world dam monitoring systems, where accuracy, speed and reliability are critical.

[2]. Impact of Arch Dam Cracking on Monitoring data

The research paper titled “Impact of Arch Dam Cracking on Monitoring Data” by Andre Conde, Miguel A. Toledo, and Eduardo Salet (2025) explores the structural and analytical challenges associated with cracks in arch dams and how these affect the interpretation of monitoring data. With the increasing reliance on monitoring systems to ensure the safety and stability of dam infrastructure, understanding the impact of physical anomalies—such as cracking—on collected data is essential. This study specifically investigates the role of hydrostatic pressure and its influence on sensor readings, with a focus on the unique structural behavior of arch dams. The proposed methodology includes a detailed analysis of how cracks alter stress distribution and pressure responses within the dam body and how these changes are reflected in sensor outputs. By integrating realistic hydrostatic pressure modeling with crack behavior analysis, the authors provide a more accurate and comprehensive view of dam health under compromised conditions.

Key parameters considered in the study include structural features of the dam, crack formation and propagation characteristics, as well as environmental factors such as water load and reservoir fluctuations. This multi-faceted parameter set enables a holistic understanding of how localized damage can distort or mislead monitoring interpretations. One of the main advantages of this approach is its ability to deliver a detailed and realistic analysis of how cracks impact dam performance and sensor feedback. The model helps engineers distinguish between normal structural responses and those caused by damage, potentially improving the accuracy of safety assessments and maintenance planning.

However, the study also highlights certain limitations. The accuracy of the results is heavily dependent on the correct placement and density of monitoring devices. Sensors that are poorly positioned may fail to capture the full extent of structural responses or may provide misleading data. Furthermore, the methodology is specifically designed for arch dams, meaning its generalizability to other types of dams—such as gravity or embankment dams—is limited. Additionally, the influence of reservoir dynamics, including varying water levels and environmental loads, introduces complexities that may affect the consistency of the analysis. Despite these limitations, the research significantly contributes to the field of dam monitoring by providing a deeper insight into the interaction between structural

damage and monitoring accuracy.

[3]. A review of detection technologies for underwater cracks on concrete dam surface

The research paper titled “A Review of Detection Technologies for Underwater Cracks on Concrete Dam Surface” by Dong Chen, Ben Huang, and Fei King, published on March 10, 2023, provides an extensive overview of modern technologies used for detecting cracks beneath water surfaces on concrete dams. As underwater inspection remains one of the most challenging aspects of dam maintenance, this paper emphasizes the importance of adopting innovative, non-invasive methods to identify structural damage that may otherwise remain undetected. The authors explore the use of underwater robots, which are equipped with cameras or sonar systems, to access submerged parts of the dam. These robotic systems allow for non- contact and non-destructive inspection, significantly reducing risk and increasing safety for inspection personnel.

The study also highlights the growing role of digital image processing and artificial intelligence in analyzing collected data. High-resolution images or sonar scans are processed using deep learning techniques, which enhance crack identification accuracy and automate detection processes. This approach reduces reliance on manual inspection and subjective visual assessments, making the process more consistent and repeatable. The parameters discussed in the paper largely pertain to underwater detection technologies, including sensor resolution, environmental adaptability, and algorithmic performance in varying aquatic conditions.

One of the key advantages of these technologies is their ability to function efficiently without physical contact, which is particularly beneficial in fragile or hard-to-reach dam areas. Additionally, the use of AI- driven analysis improves the accuracy and speed of detection, which can lead to faster decision-making in dam maintenance and risk assessment. However, the study also points out several limitations. The underwater environment is often murky, turbulent, or filled with debris, which can hinder visibility and sensor performance. Moreover, processing high volumes of data collected by robots requires significant computational power and optimized algorithms to ensure timely and accurate analysis. These computational challenges, along with potential hardware limitations in underwater robotics, can constrain the real-time applicability of these methods.

Despite these challenges, the paper concludes that combining robotics with intelligent image processing offers a promising future for underwater dam inspection. The integration of such

technologies has the potential to enhance safety, reduce inspection time, and provide more reliable data for structural health monitoring in submerged conditions.

[4]. Intelligent Detection Method for Concrete dam surface Cracks based on two stage transfer learning

The research article titled “Intelligent Detection Method for Concrete Dam Surface Cracks Based on Two- Stage Transfer Learning” by Jianyuan Li, Xiaochun Lu, Ping Zhang, and Qingquan, published on May 31, 2023, explores an advanced machine learning approach to enhance the detection of cracks on concrete dam surfaces. This study introduces a novel framework combining two-stage transfer learning with a multilayer parallel residual attention network to tackle challenges in crack identification, especially in cases with limited labeled data or complex environmental conditions. The authors present a deep learning-based solution that significantly improves the reliability of automated crack detection by leveraging knowledge from related domains and refining it for dam-specific applications.

The two-stage transfer learning approach allows pre-trained models to adapt to new datasets with minimal retraining, which is highly beneficial when acquiring large-scale, labeled dam surface crack datasets is difficult. The incorporation of a multilayer parallel residual attention mechanism further strengthens the model’s ability to focus on crack regions while minimizing background noise or irrelevant features. This results in sharper crack edge detection, improved localization accuracy, and a reduction in false positives. The parameters explored in this study include various deep learning model settings such as learning rate, layer depth, filter size, and attention weights, all optimized to enhance performance in the target domain.

This method boasts several significant advantages. Firstly, it delivers high accuracy in identifying fine cracks, which are often difficult to detect using traditional methods. Secondly, the edge refinement provided by the attention mechanism ensures that the crack boundaries are more accurately captured, which is crucial for structural health monitoring. Lastly, by utilizing transfer learning, the model reduces computation costs during training, making it more accessible for real- world deployment on limited hardware or with constrained datasets.

Despite these strengths, the paper also outlines key limitations. The performance of the proposed method is influenced by the quality and diversity of the datasets used. Dataset constraints—such as lack of variability in lighting conditions, crack types, or dam surface textures—can hinder

generalization to unseen environments. Moreover, environmental interference such as water stains, algae, or lighting variations may still confuse the model in some cases. Handling edge cases, like overlapping cracks or highly eroded regions, also remains a challenge due to the complexity of distinguishing these features from actual damage. In conclusion, this research provides a promising direction in applying intelligent deep learning techniques to structural monitoring, demonstrating that with careful model design and optimization, concrete dam crack detection can be both efficient and accurate, even under practical constraints.

[5]. A Digital Twin dam and Watershed Management

The research paper titled “A Digital Twin Dam and Watershed Management Platform” by DongSoon Par and Hojun You, published on June 1, 2023, introduces a cutting-edge approach that utilizes Digital Twin technology to revolutionize dam and watershed monitoring and management. The proposed platform integrates Geographic Information System (GIS)-based geospatial data, flood response mechanisms, water resource planning, and AI-powered CCTV surveillance to build a real-time, dynamic digital replica of the physical dam and its surrounding watershed. This holistic integration enables a proactive and predictive decision-making framework for water resource authorities and infrastructure operators.

The core concept of the platform is the Digital Twin, a virtual model that mirrors the physical state and behavior of the dam infrastructure. It continuously receives data from various sensors, surveillance systems, and hydrological models to update and simulate possible scenarios such as flood events, water level fluctuations, and structural stress. The platform incorporates AI algorithms for image recognition and anomaly detection through CCTV monitoring, ensuring prompt identification of structural threats or abnormal activities. Additionally, it leverages geospatial mapping and data analytics for long-term planning, climate adaptation, and sustainable watershed governance.

Key parameters utilized include those associated with digital twin modeling (such as data synchronization intervals, simulation fidelity, sensor coverage), watershed hydrology (precipitation, runoff, infiltration), and AI detection thresholds (e.g., for identifying cracks or debris flows via CCTV). These are optimized to provide reliable insights and facilitate accurate modeling of real-world conditions. The advantages of this system are significant. It supports enhanced decision-making by providing actionable insights based on live data and simulations. The use of AI-enhanced monitoring ensures high detection accuracy and reduces human error. Its scalability allows the platform to be adapted to dams of varying sizes and complexities, making it suitable for both large-scale national water systems and smaller regional installations. Furthermore, the integration of real-

time and historical data allows for efficient and anticipatory monitoring, which is crucial in emergency response planning, such as during flood events. Despite its promising features, several limitations are acknowledged. Weather conditions can affect the accuracy of AI-powered visual monitoring systems, especially in scenarios with heavy fog, rainfall, or low visibility. The system also demands high computational resources for real-time simulation, data processing, and 3D visualization, which may challenge less-equipped agencies or regions with limited infrastructure. Additionally, real-time data integration and decision support in high-stakes situations (e.g., flood surges) require robust and redundant communication systems to ensure reliability under pressure.

In summary, this paper presents an innovative step forward in intelligent water infrastructure management through the use of digital twins, providing a flexible, intelligent, and responsive system that bridges physical and digital domains for safer and more efficient dam and watershed operations.

[6]. Automatic Concrete Dam Deformation Prediction Model Based on TPE-STL-LSTM

The research paper titled “Automatic Concrete Dam Deformation Prediction Model Based on TPE-STL- LSTM” by Sihan Song, Qiuqing Zhou, Tao Zhang, and Yintao Hu, published on June 1, 2023, presents an innovative machine learning framework for predicting dam deformation with high temporal accuracy and automation. The method integrates Tree-structured Parzen Estimator (TPE), Seasonal-Trend decomposition using LOESS (STL), and Long Short-Term Memory (LSTM) networks to form a robust time-series prediction pipeline. This approach aims to overcome challenges in capturing nonlinear patterns and seasonal variations in dam deformation data, which are crucial for infrastructure safety and preventive maintenance.

The proposed model begins by using STL to separate the seasonal, trend, and residual components of the dam deformation dataset, enabling the learning model to focus on simplified and more structured sub- signals. These components are then processed by an LSTM neural network, which is especially effective in modeling long-range dependencies and temporal sequences. The TPE algorithm is employed for hyperparameter optimization, ensuring that the LSTM is trained under the most suitable conditions for each dam scenario.

Key parameters in the model include the number of LSTM layers, sequence lengths, STL window sizes, and optimization settings guided by the TPE algorithm. These are carefully tuned to maximize prediction accuracy while minimizing error propagation across forecasting intervals. Real-world dam datasets are used to validate the model's performance, adding credibility to its applicability in

operational environments.

The primary advantages of this approach include high prediction accuracy, thanks to the decomposition of complex signals into more manageable components and the learning capacity of LSTM networks. Additionally, the system offers automated data decomposition, reducing manual intervention and making the framework scalable to different dams and geographical conditions. The real-world validation enhances its relevance for practical deployment in safety-critical monitoring systems.

However, there are some limitations. Potential overfitting remains a concern, especially when dealing with limited historical data or non-stationary environmental conditions. The model's generalizability may also be limited across dams with drastically different geotechnical properties or sensor configurations. Moreover, the integration of STL and deep learning models introduces computational complexity, posing challenges for real-time implementation, particularly in low-resource or remote settings.

In conclusion, this paper contributes a technically sound and data-driven methodology for dam deformation prediction, combining statistical decomposition with deep learning and optimization. While promising in accuracy and automation, its real-time utility and scalability require further enhancements.

2.2 COMPARISON TABLE

PAPER TITLE	AUTHORS	PROPOSED METHOD	ADVANTAGES	DISADVANTAGES
A review of detection technologies for under water cracks on concrete dam surface	Dong chen Ben Huang, Fei King	Use of under water robot equipped with cameras for crack detection	Higher efficiency, Enhanced accuracy, non contact & non destructive	Data dependency, Cost effective
A digital twin Dam and watershed management platform	DoogSonpr, Hojun You	Digital twin technology, Flood response and water mangement	Enhanced decision making, scalability	High initial costs Complex implementation
Automatic concrete dam deformation prediction model based on TPE-STL-LSTM	Si Hanog, Qiu Jing, Zhou, Ta o Zhang Yintao H	Seasonal and trend decomposition using loess	High prediction accuracy, Real world validation	High computational cost, complex model structure
Intelligent Detection method for concrete dam surface cracks based On two stage Transfer learning	Jianyuan Li , Xiaochun Lu, Ping Zhang, Qingquan M	Two stage transfer learning	High accuracy, Robustness, Reduced cost	Increased model complexity, Longer training time

IoT Based crack detection and protection for dams	Bindu shula ,Deeptip, Gowthamivittal, Mani Deeika JM	Microcontroller & sensors, Cloud integration	Real time monitoring, cost effective	Dependence on internet connectivity, Powersupply issues, Sensor limitations
Ensemble of Deconvolutin a lneural networks for pavemen t crack detection	Zhun fan,C hong li, ingchen, Paola Di mascio, Guijie zhu	Ensemble learning with C NNs, Morpho logical crack measurement	High detection acuracy, Handel complex cracks	Computationally expensive, Limited to static images

2.3 SUMMARY

Chapter 2 presents a comprehensive review of recent advancements in dam monitoring and safety technologies. Various studies explore intelligent systems integrating optimization algorithms, AI, and sensor-based monitoring for improved accuracy and real-time analysis. Techniques such as the ABC-SVM hybrid, deep learning-based crack detection, and digital twin modeling demonstrate how automation enhances prediction, detection, and decision-making in dam management. The review highlights innovations in underwater inspection, deformation prediction, and smart data processing for structural health monitoring. Despite the progress, challenges remain in data reliability, environmental variability, computational demands, and system generalization. Overall, the chapter emphasizes that integrating AI, IoT and predictive modeling holds great potential for ensuring dam safety, efficiency and sustainability.

CHAPTER- 3

PROPOSED WORK

3.1 BLOCK DIAGRAM

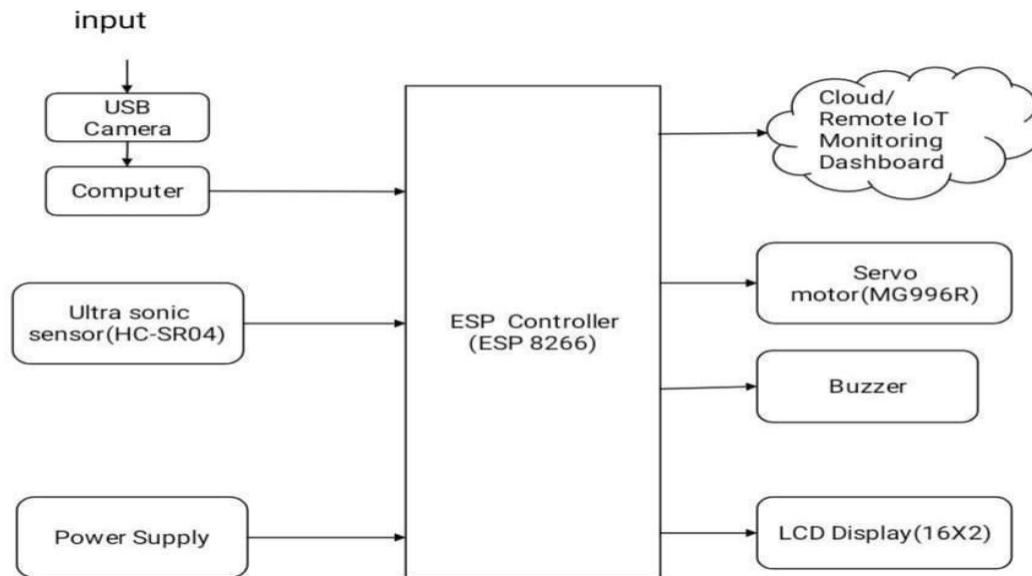


Fig 3.1 Block Diagram

The Fig 3.1 block diagram represents an IoT-based dam monitoring system that detects **cracks** and monitors the **water level** using sensors and cloud connectivity. The **ESP8266 microcontroller** acts as the main control unit, interfacing with all input and output devices. A **USB camera** connected to a **computer** captures images of the dam surface to detect cracks using image processing techniques. The computer analyzes the data and sends the results to the ESP8266 for further processing. The **ultrasonic sensor** measures the water level in the dam reservoir by calculating the distance between the sensor and the water surface. The power supply provides the necessary operating voltage to all components in the system. Based on the sensor readings, the ESP8266 triggers various output devices: a **servo motor** can be used to control dam gates automatically when water exceeds a safe level, and a **buzzer** alerts operators about abnormal conditions such as cracks or high water levels. The **LCD display** shows real-time parameters like water level and system status locally. Additionally, the ESP8266 uploads data to a **cloud or remote IoT monitoring dashboard**, enabling authorities to monitor dam safety and conditions remotely in real time. This integrated system enhances dam safety by providing early warnings and continuous monitoring.

3.2 HARDWARE COMPONENTS

- ESP 8266 Controller
- Ultrasonic Sensor (HC-SR04)
- Servo Motor (SG90 or MG996R)
- Buzzer
- LCD Display (16x2 with I2C module)
- Power Supply
- Resistors
- Bread board
- Jumper Wires
- USB cable
- Laptop

3.3 SOFTWARE REQUIREMENTS

- Arduino IDE
- MATLAB

3.2 HARDWARE DESCRIPTION

3.2.1 ESP8266 Controller:



Fig.3.2.1 ES8266 Controller

The Fig 3.2.1 **ESP8266** is a low-cost, Wi-Fi-enabled microcontroller developed by **Espressif Systems** that is widely used in **Internet of Things (IoT)** applications. It integrates a **Tensilica L106 32-bit RISC processor**, operating at clock speeds up to **80 or 160 MHz**, providing efficient processing capability for small embedded systems. The ESP8266 comes with **built-in Wi-Fi connectivity**, allowing it to connect directly to wireless networks without the need for an external module, making it ideal for smart home devices, remote monitoring, and automation systems. It features **GPIO pins**, **UART**, **SPI**, **I2C**, and **ADC interfaces**, which enable it to interact with sensors, actuators, and other peripherals. The module operates on **3.3V power supply** and supports various modes such as **station mode**, **access point mode**, and **both combined**, giving it flexibility in networking. It can be programmed using platforms like **Arduino IDE**, **NodeMCU (Lua)**, or **MicroPython**, making it beginner-friendly and popular among hobbyists and developers. The ESP8266 also supports **TCP/IP protocol stack**, allowing seamless internet communication. Overall, its **compact size**, **low power consumption**, and **affordable cost** have made the ESP8266 one of the most widely used microcontrollers for IoT and wireless communication projects.

In conclusion, the **ESP8266 microcontroller** represents a significant advancement in embedded Wi-Fi technology, offering a perfect balance of **cost**, **performance**, and **ease of use**. Its integration of processing power, wireless connectivity, and flexible I/O options within a single chip makes it an ideal choice for building smart, connected devices. Whether for educational use, prototyping, or commercial product development, the ESP8266 continues to play a vital role in shaping the future of IoT by making wireless communication simple, efficient, and accessible to everyone.

3.2.2 Ultrasonic sensor (HC-SR04) :



Fig.3.2.2 Ultrasonic sensor(HC-SR04)

The Fig 3..2.2 Ultrasonic sensor (HC-SR04) is a popular and affordable device used for measuring distances with high accuracy. It operates by emitting an ultrasonic sound wave through its "trigger" pin and then measuring the time it takes for the echo to return to its "echo" pin after reflecting off an object. Using the speed of sound, the sensor calculates the distance to the object, typically within a range of 2 cm to 400 cm, with an accuracy of about 3 mm.

The HC-SR04 requires a 5V power supply and is commonly used in robotics, obstacle avoidance systems, and DIY electronics projects for detecting objects and measuring distances. Its simple 4- pin interface (VCC, Trigger, Echo, GND) makes it easy to integrate with microcontrollers like Arduino, ESP32, and Raspberry Pi.

3.2.3 Servo Motor(MG996R) :



Fig.3.2.3 Servo Motor

The Fig 3.2.3 Servo Motor MG996R is a high-torque, standard-sized servo motor widely used in robotics, RC vehicles, and other automation projects. Manufactured by TowerPro, the MG996R is an upgraded version of the popular MG995, offering improved performance and reliability. It features metal gears, providing greater strength, durability, and resistance to wear under heavy loads. The servo operates typically at 4.8V to 7.2V, delivering a torque of about 9.4 kg-cm at 6V, and has a rotation speed of approximately 0.17 seconds per 60 degrees at the same voltage. Its control system uses PWM (Pulse Width Modulation) signals, allowing for precise angular positioning usually up to about 120 to 180 degrees. Compact, tough, and relatively affordable, the MG996R is an excellent choice for hobbyists and professionals needing reliable motion control.

3.2.4 Buzzer:



Fig.3.2.4 Buzzer

The Fig 3.2.4 Buzzer In a Dam Crack and Water Level Monitoring System, a buzzer serves as a critical alert component, providing immediate audible warnings when dangerous conditions are detected. When sensors identify cracks exceeding safe thresholds or when the water level rises beyond critical limits, the control system activates the buzzer to alert nearby operators and maintenance personnel. This rapid notification helps ensure quick responses, preventing potential structural failures or flooding. Typically, piezoelectric buzzers are used due to their low power consumption, loud sound output, and ease of integration with microcontrollers or other control units. The buzzer's simple yet effective functionality makes it an essential safety feature in monitoring systems designed to protect large-scale infrastructure like dams.

3.2.5 LCD Display(16x2):

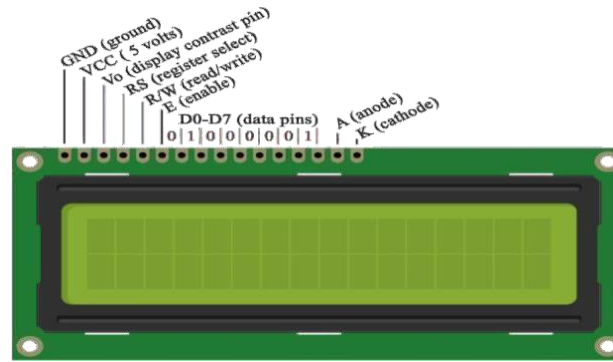


Fig.3.2.5 LCD Display

The Fig 3.2.5 LCD Display is a 16x2 character LCD module that can display up to 32 characters at once on its screen. It is based on the HD44780 driver from Hitachi. The JHD162A has 16 pins and can be operated in 4-bit mode (using only 4 data lines) or 8-bit mode (using all 8 data lines). The LCD display Module is built in a LSI controller, the controller has two 8-bit registers, an instruction register (IR) and a data register (DR). The display is 80mm x 36mm in size and the display area is 65mm x 14mm. Equipped with a 16x2 character matrix, the JHD162A LCD Display offers a clear and readable interface for conveying information to users. Each character cell can display alphanumeric characters, symbols, and custom graphics, enabling versatile representation of textual and graphical data relevant to the grading process. The high contrast and backlighting features of the LCD module ensure optimal visibility under various lighting conditions, making it suitable for indoor agricultural processing environments where lighting may vary. Whether displaying real-time sensor readings, system status updates, or sorting instructions, the JHD162A LCD Display provides operators with actionable insights to effectively monitor and control the grading system.

3.2.6 Power supply:



Fig.3.2.6 Power supply

The Fig 3.2.6 **Power supply** is an essential component used to provide a stable 5-volt output required by many electronic circuits, microcontrollers, sensors, and communication modules. It can be derived from various sources, such as batteries, USB ports, or AC-to-DC adapters, and often includes voltage regulation to ensure consistent performance even when the input voltage fluctuates. In many systems, especially in microcontroller-based projects like Arduino, ESP32, or sensor networks, a 5V power supply ensures the safe and reliable operation of all connected components. It can come in different forms, such as wall adapters, battery packs, or dedicated regulator circuits like the 7805 voltage regulator. A good 5V power supply is critical for maintaining circuit stability, protecting components from voltage spikes, and ensuring overall system reliability.

3.2.7 Resistor :



Fig.3.2.7 Resistor

The Fig 3.2.7 **Resistor** is a passive electronic component that limits or regulates the flow of electric current in a circuit. It provides resistance, measured in **ohms (Ω)**, to oppose current flow. Resistors are fundamental to almost every electronic circuit, as they help control voltage and current, divide voltages, bias active devices, and protect components from excessive current.

3.2.8 Bread Board:



Fig.3.2.8 Bread Board

The Fig 3.2.8 Bread Board is a rectangular, plastic device used to prototype electronic circuits without soldering. Its name originates from its historical use, where electronic components were mounted on literal wooden breadboards. Modern bread boards, however, are sophisticated tools designed to facilitate temporary connections between components and wires, making them essential for electronics experimentation and learning. Breadboards are essential tools in electronics, bridging the gap between theoretical design and practical implementation. They offer a versatile platform for exploring circuits, enabling rapid iteration and learning.

3.2.9 Jumper Wires :



Fig.3.2.9 Jumper Wires

The Fig.3.2.9 Jumper Wires seamlessly integrate with the Arduino Uno microcontroller and peripheral devices, serving as the primary means of communication and signal routing between various hardware modules. Whether connecting sensors for data acquisition, interfacing actuators for control, or linking display modules for user feedback, jumper wires play a crucial role in establishing reliable electrical connections that enable the seamless operation of the grading system. Through careful organization and labeling of jumper wire connections, operators can ensure clarity and ease of maintenance, simplifying troubleshooting and system maintenance tasks throughout the lifecycle of the automated grading system.

3.2.10 USB cable:



Fig.3.2.10 USB cable

The Fig3.2.10 USB cable facilitates bidirectional data transfer between the Arduino Uno microcontroller and the connected computer or programming device. Through this cable, operators can upload firmware, sketches, and code libraries to the Arduino Uno, enabling customization and configuration of the grading system's behavior and functionality. Additionally, the USB connection allows for real-time monitoring and debugging of the grading process, providing operators with insights into system performance, sensor readings, and debugging information through the Arduino Integrated Development Environment (IDE) or serial communication tools.

3.2.11 Power supply adaptor :



Fig.3.2.11 Power supply adaptor

The Fig.3.2.11 the Power Supply Adapter, rated at 12V and 2A, is a critical component in the automated grading system for agricultural products, providing the necessary electrical power to drive and operate various hardware components. With its output voltage of 12 volts and current capacity of 2 amps, this power adapter delivers a total power output of 24 watts, ensuring sufficient power supply for the operation of the conveyor belt, motor drivers, microcontrollers, sensors, and other peripheral devices within the grading system. Equipped with a standard DC barrel connector or terminal block interface, the power supply adapter offers compatibility and ease of integration with the electrical infrastructure of the grading setup.

3.2.12 Laptop:



Fig.3.2.12 Laptop

The Fig 3.2.12 Laptop plays a crucial role primarily during the development and testing phases. It is used to program the Arduino Uno via the Arduino Integrated Development Environment (IDE). Using a USB cable, the laptop allows the user to write, compile, and upload the control code (sketch) to the microcontroller. Additionally, it serves as a tool for real-time debugging and monitoring through the serial monitor feature of the Arduino IDE, where sensor data (such as readings from the LDR and ultrasonic sensor) can be observed and analyzed. The laptop also provides power to the Arduino during testing.

3.3 SOFTWARE DISCRIPTION

3.3.1 ArduinoIDE:



Fig 3.3.1 Logo of Arduino IDE

The Fig 3.3.1 Logo of Arduino IDE, Integrated Development Environment (IDE) is a software platform that allows users to write, compile, and upload code to Arduino microcontroller boards. In applications like dam crack and water level indicators, the Arduino IDE plays a crucial role in programming the sensors and microcontrollers that monitor and manage these systems. For a dam crack indicator, Arduino can be used to interface with sensors like strain gauges or ultrasonic sensors to detect shifts or cracks in the dam structure. In water level indicator systems, Arduino can process data from ultrasonic sensors or float switches to monitor the water level in reservoirs or tanks. Using the Arduino IDE, developers can easily write and upload code that processes the sensor data, triggers alarms, or activates other systems based on the real-time measurements, ensuring safety and efficient operation of these critical infrastructure systems.

3.3.2 MATLAB:



Fig3.3.2 Logo of MATLAB

The Fig 3.3.2 Logo of MATLAB (Matrix Laboratory) is a high-level programming and numeric computing software developed by MathWorks. It is widely used for mathematical modeling, algorithm development, data analysis, and simulation. MATLAB provides a powerful environment for working with matrices, plotting graphs, and implementing complex mathematical equations easily. It supports various toolboxes for specialized applications like image processing, control systems, signal processing, and machine learning. Engineers, scientists, and researchers use MATLAB for designing and testing real-world systems efficiently. Its user-friendly interface and extensive library make it a versatile tool in both education and industry.

3.4 SUMMARY

Chapter 3 explains the proposed IoT-based dam monitoring system designed to detect structural cracks and monitor water levels in real time. The system uses an ESP8266 microcontroller as the main control unit, interfaced with sensors, actuators, and a cloud platform for data transmission and remote monitoring. An ultrasonic sensor measures water levels, while a USB camera and MATLAB- based image processing detect cracks on the dam surface. The system triggers alarms and operates a servo motor to control dam gates when critical thresholds are reached. Key hardware components include the ESP8266, ultrasonic sensor, servo motor, buzzer, LCD display, resistors, and power supply. Jumper wires, breadboards, and a laptop are used for circuit setup, programming, and testing. The Arduino IDE is employed for coding and uploading microcontroller programs, while MATLAB handles image analysis for crack detection. This combination of hardware and software ensures efficient, automated monitoring with real-time alerts. The design prioritizes safety, scalability, and reliability, enabling authorities to respond quickly to potential hazards. Overall, the proposed system enhances dam management through continuous, smart, and data-driven monitoring.

CHAPTER- 4

IMPLEMENTATION

4.1 Hardware Implementation

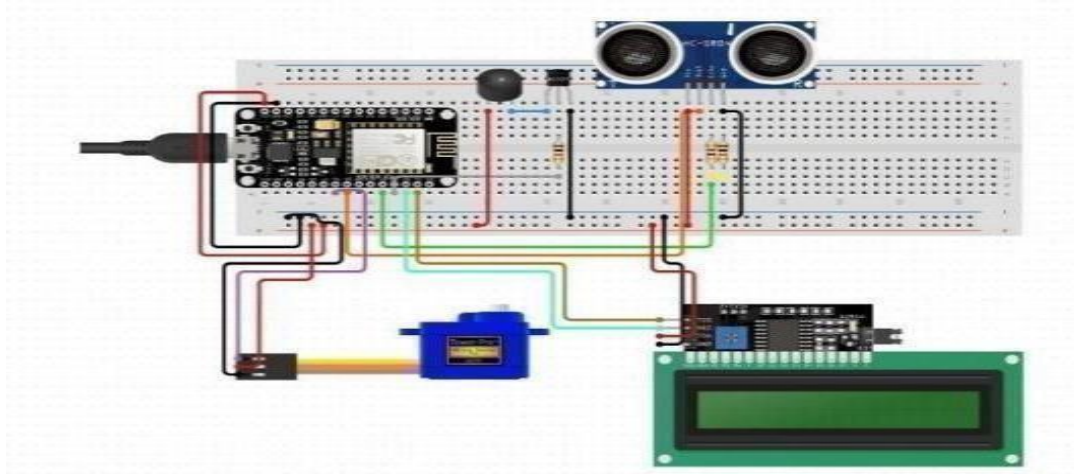


Fig.4.1 Circuit Diagram of Water Level Indicator

The Fig 4.1 Circuit Diagram of Water Level Indicator shows proposed working model for water level indication and automatic gate control in a dam is designed to enhance safety, efficiency, and automation in water management systems. In this project, an **“ultrasonic sensor”** is deployed inside the dam structure to continuously monitor the water level. The sensor works on the principle of transmitting ultrasonic sound waves and measuring the time taken for the echo to return after hitting the water surface. Based on this time, the distance between the sensor and water surface is calculated, which is then converted into the actual water level measurement. This non-contact measurement method makes the ultrasonic sensor highly suitable for dam applications, as it avoids wear and tear and works effectively even in harsh environmental conditions. The readings from the sensor are transmitted to the ESP8266 controller module, which acts as the brain of the system. The controller processes the data, compares it with predefined water level thresholds, and makes real-time decisions regarding the operation of dam gates.

The ESP8266 microcontroller plays a critical role by not only processing the sensor data but also enabling wireless connectivity for remote monitoring. Whenever the water level remains within safe limits, the system continues to monitor without taking further action. However, when the water level starts rising beyond a critical threshold, the microcontroller automatically triggers the gate-opening mechanism. The dam gates are connected to electric actuators, which are powerful devices capable

of converting electrical signals into mechanical movement. Upon receiving a command from the controller, the actuators activate and open the gates of the dam. In this project, three gates are integrated into the model, and they open automatically when the water level crosses the danger point, thereby releasing excess water to prevent overflow, flooding, or structural stress on the dam. The actuator ensures smooth and controlled opening of gates to maintain safe water flow.

To implement a dam crack detection system using an ESP8266 microcontroller, laptop camera, and the Blynk app, we need to integrate computer vision with IoT communication for real-time monitoring. First, the laptop camera is used to continuously capture images of the dam structure. This image feed is processed locally on the laptop using Python and OpenCV, where a trained crack detection algorithm (based on edge detection or a pre-trained deep learning model like CNN) analyzes the frames for cracks. If a crack is detected beyond a datasets the detection result is communicated to the ESP8266, which acts as a Wi-Fi-enabled IoT module. The ESP8266 receives the data via serial communication (USB or UART) from the laptop or over Wi-Fi using a lightweight protocol such as MQTT or HTTP. Once it receives the alert or crack data, the ESP8266 pushes this information to the Blynk IoT platform using its Wi-Fi capabilities. In the Blynk app, a user-friendly interface can be created to display the crack status, image snapshots, timestamps, and alert messages. Additionally, features like email or push notifications can be configured through Blynk. This system provides a cost-effective and scalable solution for remote dam monitoring, where the laptop handles intensive image processing, and the ESP8266 serves as a communication bridge to the cloud, enabling real-time alerts and visualization on mobile devices via Blynk.

4.2 Software Implementation

4.2.1 Flow chart of water level indicator

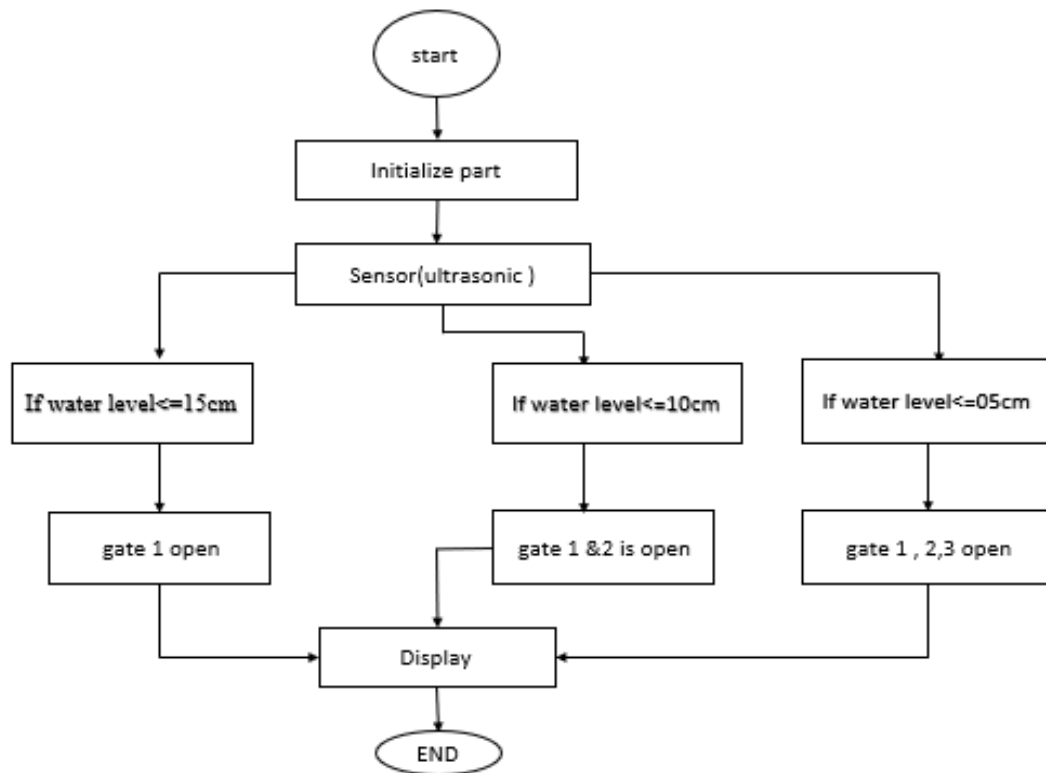


Fig.4.2.1 Flow chart of water level indicator

The Fig 4.2.1 Flow chart of water level indicator represents a water level monitoring and control system. The process begins with the initialization of system components, followed by the activation of a sensor that measures the water level. Based on the readings, the system checks if the water level is less than or equal to 20. If this condition is met, it displays that the water level is normal and proceeds to activate a buzzer as a signal before ending the process.

If the water level exceeds 20, the system further checks whether it is less than 30. If true, it displays that Gate 1 is open and activates the buzzer. However, if the water level is 30 or more, the system indicates that both Gate 1 and Gate 2 are open and again activates the buzzer before ending.

In summary, the chart outlines a sequential control system that uses a sensor to monitor water levels and responds by updating display messages and triggering a buzzer based on defined water level thresholds.

The Arduino program is designed to measure the distance of an object using an ultrasonic sensor and respond with visual and mechanical feedback. At startup, the system initializes the LCD display, servo motor, and pins for the ultrasonic sensor and buzzer. The LCD displays a message to indicate initialization, then clears for regular operation. In the main loop, the program continuously reads the distance using the ultrasonic sensor.

1. System Initialization

Set up all hardware components (LCD, ultrasonic sensor, servo motor, and buzzer)

Steps: 1. Initialize the LCD display:

- Start communication with LCD
- Turn on backlight
- Show "Initializing..." message

Define pin modes:

- TRIG_PIN: OUTPUT (to send ultrasonic pulses)
- ECHO_PIN: INPUT (to receive echo)
- BUZZER_PIN: OUTPUT (to control buzzer)
- SERVO_PIN: Attach servo motor and initialize to 0 degrees

2. Main Loop Execution

Continuously measure distance and respond with servo movement and buzzer status Steps: 1. Measure

Distance (call getDistance() function):

- Trigger the ultrasonic sensor
- Measure the time taken for echo to return
- Convert time to distance using formula

2. Display Distance on LCD:

Show measured distance in cm on the first row

3. Distance Comparison:

- If distance is less than threshold (5 cm):
- Rotate servo to 180°
- Turn buzzer ON (write LOW to active buzzer)
- UpdateLCD: "Servo:90 Buzz\ :ON"

If distance is greater than or equal to threshold:

- Rotate servo to 180°
- Turn buzzer OFF (write HIGH to buzzer)
- UpdateLCD: "Servo:0 Buzz\ :OFF"

4. Wait 500 ms before repeating the loop getDistance() Function Algorithm

Measure distance using ultrasonic sensor

Steps: 1. Set TRIG_PIN LOW for 2 μ s (to ensure clean LOW)

2. Set TRIG_PIN HIGH for 10 μ s (send a pulse)

3. Set TRIG_PIN LOW again

4. Use pulseIn() to measure duration of the echo on ECHO_PIN

5. Calculate distance using:

$$\text{distance} = \text{duration} \times 0.0343 / 2$$

- 0.0343 cm/ μ s is speed of sound in air
 - Divide by 2 because the sound travels to the object and back
6. Return the computed distance in centimeters.

4.2.2 Flow chart of crack detection

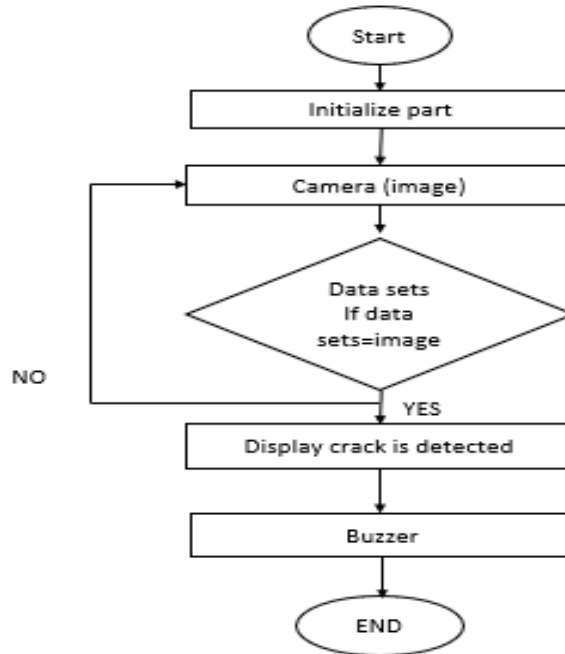


Fig.4.2.2 Flow chart of crack detection

The Fig.4.2.2 Flow chart of crack detection outlines a basic process for detecting cracks using an ESP8266 controller, a laptop camera, and a buzzer.

The process begins by initializing the system's components including the ESP8266 and the laptop camera. The system then captures an image from the laptop camera. This image is compared against a pre-existing dataset of known crack patterns. If the captured image does not match any of the crack patterns in the dataset, the process loops back to the initialization step, continuously capturing and checking for cracks. However, if a match is detected, the system displays a message indicating that a crack has been found. Concurrently, a buzzer is activated to provide an audible alert. After these actions, the system concludes its operation. This workflow is a fundamental example of an automated visual inspection system.

In MATLAB, image processing plays a crucial and foundational role in the detection of cracks in dam structures, serving as the initial and most essential stage before applying deep learning classification using architectures like ResNet50. The process begins with image acquisition, where high-resolution images of the dam surface are captured using digital cameras, drones, or specialized sensors. These images serve as the raw input data for the system. Since the quality of results in any image-based system depends heavily on the quality of the input, MATLAB provides several tools to

correct distortions, remove background noise, and improve clarity. Images are first imported into the MATLAB workspace using functions like `imread`, and then preprocessed to ensure uniformity and accuracy across the dataset. The preprocessing stage involves converting colored images to grayscale using `rgb2gray`, which simplifies computational complexity while retaining important texture and intensity information. Noise present due to lighting variations, camera vibration, or environmental factors is removed using spatial filtering techniques, such as median or Gaussian filters implemented with functions like `medfilt2` and `imgaussfilt`. These steps enhance image clarity and ensure that only the relevant surface features are retained for further analysis.

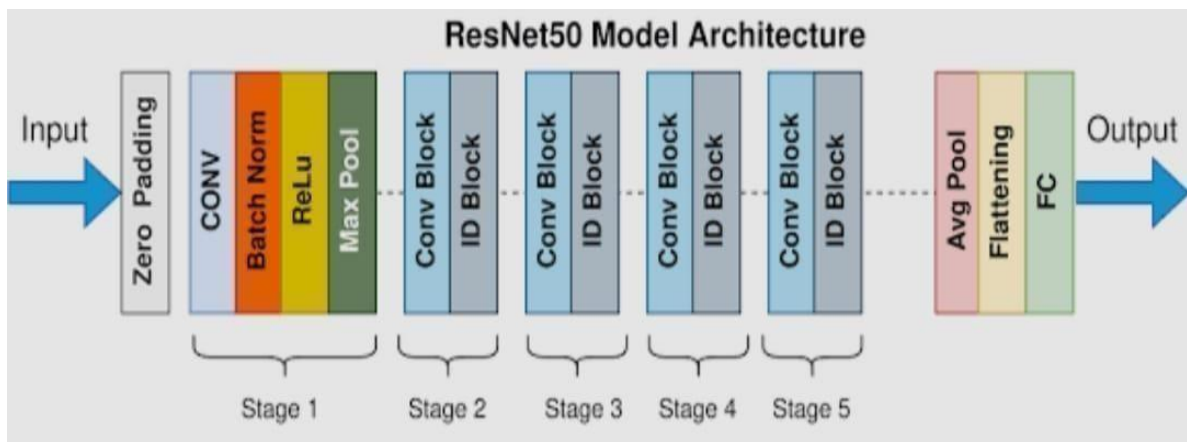


Fig.4.2.3 ResNet50 Model

The Fig.4.2.3 ResNet50 Model shows once the images are denoised, contrast enhancement is applied to highlight fine cracks that might be invisible to the naked eye. MATLAB uses functions such as `imadjust`, `histeq`, or adaptive histogram equalization (`adapthisteq`) to improve visibility by adjusting brightness and contrast levels automatically. This ensures that even faint cracks are distinguishable from the background concrete texture. After enhancement, the images undergo segmentation, which separates the cracked regions from the non-cracked areas. Segmentation can be performed using thresholding techniques like Otsu's method (`graythresh` and `imbinarize`), which converts grayscale images into binary form where cracks appear as dark lines against a bright background. For more complex textures, edge detection algorithms such as Sobel, Canny, or Laplacian filters are employed to detect the boundaries of cracks accurately. MATLAB's Image Processing Toolbox provides built-in functions like `edge` for this purpose, which helps in extracting structural outlines of cracks from the dam's surface image.

Following segmentation, morphological operations are applied to refine the detected crack regions. Functions such as `imopen`, `imclose`, `bwareaopen`, and `imdilate` are used to remove small irrelevant objects, fill gaps, and connect broken crack segments. This process enhances the continuity and

sharpness of the cracks, making them more distinct and measurable. Once the crack regions are clearly isolated, MATLAB's region analysis tools such as `regionprops` can be used to measure crack properties including length, width, and orientation. These measurable features can be stored and used for statistical analysis or further classification tasks. At this stage, the preprocessed images are ready for deep learning analysis. They are resized to match the input dimensions required by the ResNet50 architecture (typically 224×224 pixels) using the `imresize` function, ensuring that all images conform to a standardized format for model training and classification.

ResNet50, a powerful convolutional neural network (CNN), is used in MATLAB via the Deep Learning Toolbox. It is a pre-trained model that has been trained on the ImageNet dataset and is capable of learning complex features automatically. In the context of crack detection, MATLAB allows for transfer learning, where the existing layers of ResNet50 are fine-tuned using a custom dataset of dam surface images. The initial layers of ResNet50 extract low-level features such as edges, lines, and color gradients, while the deeper layers extract high-level features such as texture patterns, intensity variations, and shapes that are unique to cracks. By replacing and retraining the final classification layers, the network becomes specialized in distinguishing between cracked and "non-cracked" dam surfaces. MATLAB provides tools like `trainNetwork` and `imageDatastore` for efficient data management, training, and validation. The training process involves splitting the dataset into training and validation sets, normalizing image intensities, and optimizing network parameters to minimize classification error.

Once training is complete, MATLAB's evaluation tools such as confusion matrices (`confusionchart`), accuracy plots, and performance metrics are used to validate the model's performance. The output from the ResNet50 classifier labels each image as either containing a crack or not, and the results can be visualized using MATLAB's plotting and image display functions. The integration of image processing with deep learning in MATLAB ensures high precision, as the preprocessing enhances visual features while the deep neural network performs robust feature extraction and classification. This system not only automates the process of crack detection but also provides a reliable, repeatable, and objective method to monitor the structural health of dams. By continuously analyzing dam surface images, MATLAB-based systems can help in early crack detection, preventive maintenance, and safety assessment, ultimately reducing the risk of catastrophic structural failures. Thus, MATLAB's comprehensive environment for image processing, combined with the deep learning capabilities of ResNet50, offers a powerful and intelligent solution for real-world applications in dam crack monitoring and infrastructure health analysis.

4.3 SUMMARY

Chapter 4 focuses on the implementation of the IoT-based dam monitoring system, which integrates hardware and software components for real-time crack detection and water level management. The hardware setup uses an ESP8266 microcontroller connected to an ultrasonic sensor, servo motor, buzzer, and LCD to measure water levels and automatically control dam gates. When the water level crosses predefined thresholds, the system triggers alerts and gate movements to ensure safe operation. For crack detection, a laptop camera captures dam surface images, which are analyzed using MATLAB and OpenCV algorithms to identify cracks. The results are communicated to the ESP8266, which uploads alerts and data to the Blynk IoT platform for remote monitoring. The software implementation includes flowcharts detailing how the system processes water level and crack data, triggers outputs, and updates status displays. MATLAB plays a vital role in preprocessing images, applying filters, segmentation, and edge detection before classifying them using the ResNet50 deep learning model. The ResNet50 network, trained through transfer learning, accurately distinguishes between cracked and non-cracked surfaces. This chapter demonstrates the seamless integration of embedded systems, IoT, and artificial intelligence to create an efficient, automated, and intelligent dam safety monitoring solution.

CHAPTER – 5

RESULTS

5.1 Water Level indication

The Fig.5.1 Water level indicator model shows implementation of the water level indication and automatic gate control system in dams using an ultrasonic sensor, ESP8266 controller, and actuator has shown effective and reliable results. The ultrasonic sensor continuously monitors the water level in the dam reservoir by sending and receiving ultrasonic pulses, and the ESP8266 processes this data in real-time. The system is designed with threshold levels, where the normal, warning, and critical levels of water are defined. When the water level rises beyond the safe threshold, the ESP8266 triggers the actuator mechanism connected to the dam gates. As a result, the gates open automatically to release excess water and prevent overflow or flooding, ensuring the safety of nearby areas. The system provides accurate, non-contact measurement and eliminates manual intervention, which reduces human error and increases operational efficiency. Additionally, due to the Wi-Fi capability of the ESP8266, the data can be transmitted to a remote monitoring station or mobile application for real-time supervision and control. This project demonstrates that the integration of IoT with water level management enhances both safety and automation in dam operations. The results confirm that the system not only improves monitoring accuracy but also offers quick, automated response, which is crucial in emergency conditions.

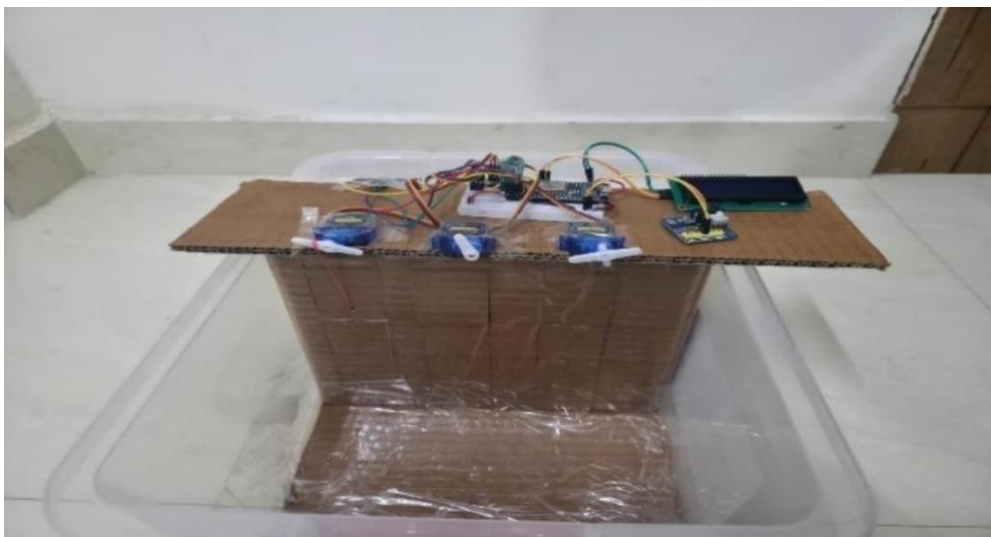


Fig.5.1.1 Water level indicator model



Fig 5.1.2 Gate 1 will open when water at low level

In the automatic dam gate control system using an ESP8266 microcontroller, ultrasonic sensor, and servo motors, water level is monitored in real-time to automate gate operation based on predefined thresholds. The system is designed with three key parameters that determine gate actions.

1. Low Water Level: The Fig.5.1.2 Gate 1 will open when water at low level shows, When the water level in the dam drops below a specific threshold, it signals a low water level condition. The system's ESP8266 controller, using a water level sensor, will detect this drop. Once this condition is met, the controller will activate a relay connected to Gate 1, causing it to open. This gate could be designed to release water from an external source or a reserve to replenish the dam. The controller keeps this gate open until the water level rises to the normal range, at which point it closes Gate 1 and transitions to the next condition.

2. Normal Water Level: The Fig.5.1.3 Gate1 and Gate2 will open when water at normal level shows condition is the desired state for the dam. When the water level is within this pre-defined range, the ESP8266 controller will keep Gate 3 closed. Instead, it will activate the relay connected to Gate 1 and Gate 2, causing it to open. This gate is the primary outlet for the dam, allowing for a steady, controlled release of water for downstream use, such as irrigation or power generation. The controller maintains this condition until the water level either drops too low or rises too high, ensuring a stable and efficient flow.



Fig 5.1.3 Gate 1 and Gate 2 will open when water at normal level

3. High Water Level (Above Normal): The Fig.5.1.4 All Gates will open when water at high level condition indicates that the dam is approaching its capacity, which could lead to flooding if not managed. When the water level sensor detects that the water has exceeded the normal range, the ESP8266 controller will open all gates. This gate is a spillway or an emergency outlet designed to release a large volume of water quickly. Opening all Gates prevents the dam from overflowing and protects the dam's structure and downstream areas from potential damage. The controller keeps this gate open until the water level returns to the normal range, at which point all gates will open.



Fig.5.1.4 All Gates will open when water at high level

5.2 PARAMETER TABLE

WATER LEVEL/DISTANCE	OPERATIONS
SERVO 1 15 TO 10 CM	GATE 1 OPEN
SERVO 2 10 TO 5 CM	GATE 1 and 2 OPEN
SERVO 3 5 TO 1 CM	GATE 1,2 and 3 OPEN

5.2 Crack Detection

The dam crack detection system, utilizing an ESP8266 controller and a laptop camera, operates on a principle of image classification. The laptop camera captures a live image of the dam's surface and sends it to a program, likely running on the laptop itself, which uses the ESP8266 for control and communication. This program then processes the image, comparing it to a pre-existing dataset. This dataset is a crucial component, divided into two categories: a positive dataset containing images of cracks on a dam surface, and a negative dataset with images of an unblemished surface.

If the algorithm determines the captured image has a high degree of similarity to the images in the positive dataset, it classifies it as containing a crack. This classification triggers two immediate results: a buzzer sounds to provide an audible alert, and a signal is sent to the Blynk app. On the app's dashboard, a message or indicator will change to display "Crack is Detected." Conversely, if the algorithm finds a strong match with the negative dataset (i.e., no crack is present), no audible alarm is given, and the Blynk app's display updates to show "No Crack." This system provides a simple yet effective method for real-time remote monitoring of structural integrity.

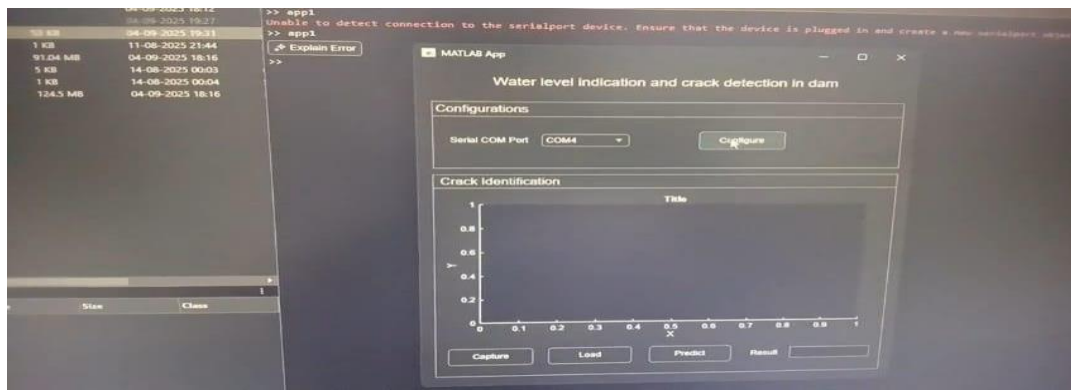


Fig.5.3.1 Capturing crack image in dam



Fig.5.3.2 Negative data set



Fig.5.3.3 Captured image matches with negative data set

The Fig.5.3.3 shows Captured image matches with negative data set , the captured image is compared with a negative dataset consisting of images without cracks or with safe surface patterns. If the captured image matches the negative dataset, the system identifies the surface as stable and free from damage. The algorithm classifies it as “No Crack” or “Safe” ensuring accurate monitoring of dam safety. This result is then transmitted through the IoT platform using the Blynk tool. On the Blynk dashboard, the status is displayed in real-time for remote monitoring. This integration improves safety, efficiency, and quick decision-making in dam maintenance.



Fig.5.3.4 Positive Data set



Fig.5.3.5 Captured image matches with positive data set

The Fig.5.3.5 Shows Captured image matches with positive data set, In the crack detection system for dams using image processing, the captured image is continuously compared with the positive dataset that contains images of cracks and damaged areas. When the captured image matches with this positive dataset, the system classifies the condition as “Crack Detected” or “Dangerous.” The image processing algorithm, supported by deep learning models, analyzes the crack’s features such as width, depth, and pattern to confirm the damage. Once confirmed, the system triggers an alert signal for immediate attention. This alert is transmitted to the Blynk IoT platform, where the real-time status “Crack Detected” is displayed. At the same time, a buzzer is activated to provide an audible warning for quick response. This smart automation ensures timely detection of structural issues, enhances safety monitoring, and prevents major dam failures.

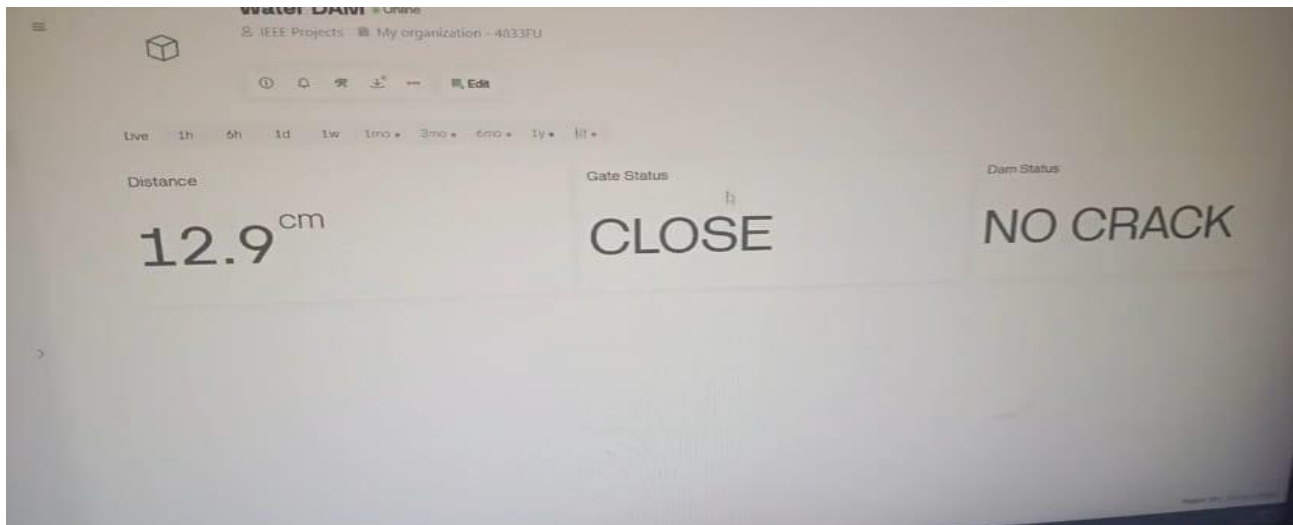


Fig.5.3.6 All Dam Parameters shown in Blynk app

The Fig.5.3.6 All Dam parameters shown in Blynk app. In a dam health monitoring system, various critical parameters are continuously measured and displayed in the Blynk app for remote supervision and safety assurance. The system primarily tracks the water level in the dam using ultrasonic or pressure sensors, ensuring that the reservoir is maintained within safe limits. Whenever the water level rises beyond the threshold, the system automatically controls the dam gate through actuators, partially or fully opening them to release excess water and prevent overflow or flooding. Additionally, the structure’s integrity is monitored using crack detection sensors like piezoelectric, strain gauge, or vibration sensors that immediately notify if any cracks or stress points are detected in the dam wall, signaling possible structural weaknesses. All this real-time data—water level, gate status, and crack detection—is transmitted wirelessly via an IoT module such as the ESP8266/ESP32 and visualized on the Blynk mobile app dashboard. This allows engineers and authorities to receive instant

notifications, monitor trends, and make quick decisions, thereby enhancing dam safety, early warning, and disaster prevention through a reliable, user-friendly interface.

5.3 SUMMARY

Chapter 5 presents the results of the **Water Level Indication and Crack Detection System for Dams** using IoT and image processing technologies. The system effectively monitors the dam's water level using an **ultrasonic sensor** and automatically controls gate operations through the **ESP8266 controller** and actuators based on predefined thresholds—low, normal, and high levels. This automation ensures timely water release and prevents overflow or flooding. The **crack detection module** employs image processing to compare captured dam surface images with positive and negative datasets. If a crack is detected, a **buzzer alert** and notification are sent to the **Blynk IoT app**, ensuring immediate attention. All parameters, including water level, gate status, and crack condition, are displayed in real-time on the **Blynk dashboard**, enabling remote supervision. The results confirm that the system enhances dam safety, reduces human error, and provides reliable, automated monitoring for effective dam management.

CHAPTER- 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The project on Water Level Indication and Crack Detection in Dams using IoT and Image Processing provides an efficient, reliable, and automated solution for dam safety monitoring. The system integrates two crucial aspects—continuous water level measurement and structural crack detection—to ensure the stability and safety of the dam structure. Using ultrasonic sensors, the water level inside the dam is constantly monitored and compared with predefined threshold levels. When the water level exceeds the safe limit, the system can automatically control the gate mechanism to release excess water, thereby preventing overflow or potential damage. Simultaneously, image processing techniques are employed to detect cracks on the dam surface using captured images, which are analyzed through trained datasets to identify any structural deformities. If the captured image matches the positive dataset indicating cracks or damage, the system classifies it as “Crack Detected” or “Dangerous,” displays the alert on the Blynk IoT platform, and activates a buzzer to ensure immediate attention. The integration of IoT allows real-time monitoring and remote access to dam conditions through the Blynk application, enabling authorities to take timely preventive measures. This project significantly reduces manual inspection efforts, minimizes human errors, and enhances the accuracy and speed of dam monitoring operations. Overall, the developed system ensures greater safety, sustainability, and efficiency in water management and infrastructure maintenance, making it a valuable tool for modern smart dam systems.

6.2 FUTURE SCOPE

- **Advanced Machine Learning Integration:** The system can be improved by integrating advanced AI and machine learning algorithms to automatically classify the severity of cracks and predict potential structural failures before they occur.
- **Real-Time Cloud Data Storage:** Future developments can include cloud-based data storage and analytics, enabling continuous monitoring, long-term data analysis, and remote access to dam safety information from anywhere.
- **Enhanced Sensor Network and Automation:** Additional sensors like pressure, vibration, and humidity sensors can be added to monitor multiple parameters, and the system can be upgraded for full automation of dam gate operations based on real-time data analysis.

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APPENDIX

CODE

```
#include <Wire.h>
#include<LiquidCrystal_I2C.h>
#include<Servo.h>

//LCDsetup(Change0x27toyourLCD's I2Caddressifneeded)
LiquidCrystal_I2C lcd(0x27, 16, 2);

//Servo
Servo myServo;
//Pins
#define TRIG_PIN D5
#define ECHO_PIN D6
#defineBUZZER_PIN D7
#defineSERVO_PIN D4

//Distancethreshold
#defineDISTANCE_THRESHOLD5//incm

voidsetup(){
  // Initialize LCD
  lcd.init();
  lcd.backlight();
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Initializing...");

  // Initialize pins
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  // Servo setup
  myServo.attach(SERVO_PIN);
  myServo.write(0); //Startat0degrees
  delay(500);
  lcd.clear();
}
```

```

void loop(){
  //Get distance
  float distance = getDistance();

  // Display distance
  lcd.setCursor(0, 0);
  lcd.print("Dist: ");
  lcd.print(distance);
  lcd.print(" cm ");

  if(distance < DISTANCE_THRESHOLD) {
    //Rotateservo to 180 degrees and turn on buzzer
    myServo.write(180);
    digitalWrite(BUZZER_PIN, LOW);

    // Update LCD
    lcd.setCursor(0, 1);
    lcd.print("Servo: 90 Buzz: ON");
  } else {
    //Rotateservo to 0 degrees and turn off buzzer
    myServo.write(0);
    digitalWrite(BUZZER_PIN, HIGH);

    // Update LCD
    lcd.setCursor(0, 1);
    lcd.print("Servo: 0 Buzz: OFF");
  }

  delay(500); //Read every 0.5 seconds
}

//Function to measure distance from ultrasonic sensor
float getDistance() {
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);

  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);

  long duration = pulseIn(ECHO_PIN, HIGH);

```



```

floatdistance=duration*0.0343/2;//Converttocm
return distance;
}

```

Crackdetection code

```

classdef ImageClassifier
    properties
        ImageDatastoreObj % Datastore for image data
        TrainingImages % Training image datastore
        TestingImages % Testing image datastore
        DeepNetwork %Pretraineddeeplearningnetwork
        FeatureLayer % Layer for feature extraction
        TrainedSVM % Trained SVM classifier
    end

    methods
        %%Constructor Method
        function obj= ImageClassifier(imageFolderPath)
            %Constructorinitializespropertiesandpreparesthe dataset
            clc;
            disp('*ImageClassificationUsing Deep Learning*');

            % Loadimagedatastore
            obj.ImageDatastoreObj = imageDatastore(imageFolderPath, 'LabelSource', 'foldernames',
'IncludeSubfolders', true);
            obj.ImageDatastoreObj.ReadFcn= @ImageClassifier.customReader;

            % Displaydataset label counts
            labelCounts=countEachLabel(obj.ImageDatastoreObj);
            disp('Image count per category:');
            disp(labelCounts);

            %Partitiondatasetintotraining andtestingsets
            [obj.TrainingImages,obj.TestingImages]=splitEachLabel(obj.ImageDatastoreObj,0.90, 'randomize');

            % Loadpretrained network
            obj.DeepNetwork =resnet50;
            obj.FeatureLayer = obj.DeepNetwork.Layers(end - 2, 1).Name;
            disp(['UsingResnet 50,extractingfeaturesfromlayer: 'obj.FeatureLayer]);
        end
    end
end

```

```

%% Feature Extraction Method
function features=extractFeatures(obj,imageSet)
    batchSize = 64;
    networkInputSize=obj.DeepNetwork.Layers(1).InputSize;
    augmentedImages=augmentedImageDatastore(networkInputSize, imageSet, 'ColorPreprocessing',
'gray2rgb');
    features=activations(obj.DeepNetwork, augmentedImages,obj.FeatureLayer,...
        'MiniBatchSize',batchSize,'OutputAs','columns','ExecutionEnvironment','auto');
end

%%TrainSVMClassifierMethod
function obj= trainClassifier(obj)
    disp('Training SVM Classifier...');
    trainingFeatures=obj.extractFeatures(obj.TrainingImages);
    trainingLabels = obj.TrainingImages.Labels;
    obj.TrainedSVM=fitcecoc(trainingFeatures,trainingLabels,...
        'Learners','svm','Coding','onevsall','ObservationsIn','columns','Verbose',2);
end

%%TestandEvaluateClassifierMethod
function evaluateClassifier(obj)
    disp('Evaluating Classifier...');
    testFeatures=obj.extractFeatures(obj.TestingImages);
    predictedLabels=predict(obj.TrainedSVM,testFeatures,'ObservationsIn','columns');
    trueLabels = obj.TestingImages.Labels;

    % Confusion matrix
    confusionMatrix=confusionmat(trueLabels,predictedLabels);
    confusionchart(trueLabels, predictedLabels);
    disp('Confusion Matrix (Raw Counts):');
    disp(confusionMatrix);

    %Normalizeconfusionmatrix
    normalizedMatrix=bsxfun(@rdivide,confusionMatrix,sum(confusionMatrix,2));
    disp('Confusion Matrix (Percentage Form):');
    disp(normalizedMatrix);

    %Calculateanddisplayclassificationaccuracy
    accuracy = mean(diag(normalizedMatrix));
    disp(['OverallClassification Accuracy=' num2str(accuracy)]);

```

```

end

%% Save Model Method
function saveModel(obj)
disp('Saving models...');
    trainedSVMClassifier = obj.TrainedSVM;
    deepNetwork = obj.DeepNetwork;
    save('trainedSVMClassifier','trainedSVMClassifier');
    save('deepNetwork', 'deepNetwork');
    disp('Modelssavedsuccessfully.');
```

end

```

end

methods(Static)
    %% Custom Image Reader Function
    function img=customReader(fileName)
    onState= warning('off','backtrace');
    c=onCleanup(@()warning(onState));
        data = imread(fileName);
        data=data(:, :,min(1:3,end));
        img=imresize(data,[256 256]);
    end
end
end
```

SDG's

Based on the project's focus on crack detection and water level indication in dams, here are the most relevant Sustainable Development Goals (SDGs):

1.SDG 6: Clean Water and Sanitation - This goal is directly related to the project's focus on water level monitoring in dams, which is crucial for managing water resources and ensuring access to clean water.

2.SDG 9: Industry, Innovation, and Infrastructure - The project involves developing and implementing technology for monitoring dam safety, which falls under the development of resilient infrastructure

3.SDG 11: Sustainable Cities and Communities - Dams play a vital role in community safety and well-being by preventing floods and ensuring water availability. The project contributes to this goal by enhancing dam safety through crack detection and water level monitoring.

4.SDG 13: Climate Action- Dams are important for water storage and can play a role in climate change adaptation. Monitoring their safety and efficiency is essential in the context of climate change and its impacts on water resources.

VISION AND MISSION

Vision and Mission statement of the college

Vision:

“To be a Leader in Imparting Value based Education in Technology, Management and Research for the benefit of Society”

Mission:

M1	To Use Modern Pedagogical methods to Promote an Ecosystem for Learning and Innovation
M2	To Undertake Entrepreneurial, Managerial and Technical Skill Development Programmes for Academic Institutions and Industries
M3	To Enhance Institute Industry Interaction through Collaborative Research and Consultancy
M4	To Relentlessly pursue Professional Excellence with Ethical and Moral values.

Vision and Mission statement of the Electronics and Communication Engineering Department:

Vision:

“To develop outstanding Electronics and Communication Engineers to meet the ever changing social and Technological needs of the Society”

Mission:

M1	To provide State-of-the-Art infrastructure in Electronics and Communication Engineering.
M2	To disseminate strong theoretical and practical exposure to meet the emerging trends in the industry.
M3	To promote a freethinking environment with innovative teaching-learning pedagogy.
M4	To develop value based socially responsible professionals for the betterment of the Society.

PROGRAM OUTCOMES

NATIONAL BOARD OF ACCREDITATION

Annexure-II: Program Outcomes (POs)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for
i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Program Specific Outcomes (PSO's)

PSO1	Develop the components for analog and digital systems, communication systems, control, and signal processing systems using acquired knowledge of basic skills and various design tools.
PSO2	Formulate the solution for interdisciplinary problems through acquired programming knowledge in the respective domain by complying with real-time constraints.

Published Paper

International Conference on Recent Trends in Technology – 2025 - 2nd May 2025

REVIEW PAPER ON MONITORING STRUCTURAL INTEGRITY AND WATER LEVELS AND ADVANCED TECHNIQUES FOR CRACK DETECTION

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ABSTRACT:

This study presents a unified approach to enhancing infrastructure monitoring and water resource management using advanced digital technologies. The first part focuses on automated pavement crack detection through an ensemble of convolutional neural networks (CNNs) without pooling layers. This method improves spatial accuracy and, combined with structured prediction and probability fusion, delivers high performance in detecting and measuring various crack types. Morphological analysis and skeleton extraction further enable precise crack width and length estimation, validated on benchmark datasets with superior results compared to traditional methods. The second part introduces a digital twin-based platform for dam and watershed management, developed for Korea's Sumjin River basin. The platform integrates high-resolution 3D geospatial modeling using aerial LiDAR and drone photogrammetry with real-time data on dam operations, rainfall, and river conditions. It supports smart decision-making through AI-powered water level prediction, flood simulation, and levee safety evaluation. The platform also incorporates automated drones and AI-driven CCTV monitoring for enhanced surveillance and response. Together, these solutions demonstrate how deep learning and digital twin technologies can modernize civil infrastructure, enabling accurate monitoring, proactive planning, and data-driven decision-making for safer and more sustainable systems.

KEYWORDS:

Automated pavement crack detection, Deep learning, Ensemble network, Convolutional neural network, Segmentation, Morphological, Dam, River management, Watershed, Water resource.

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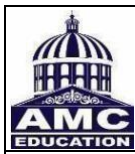
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PROJECTWORK SELF EVALUATION FORM						
Name	NAVEEN B	USN	1AM22EC072			
Program	Bachelor of Engineering in Electronics and Communication					
S.No	To what extent did each of the following you contribute to your project	To a very great extent	To a great extent	To a moderate extent	To some extent	Not at all
1.	Have you applied the knowledge of mathematics, science and engineering concepts(PO1,Pso1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
2.	Have you analyzed the system of hardware, software and its interaction (PO2,PSO1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
3.	Design the systems, components or processes to meet needs(PO3,PSO1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
4.	Have you used the modern tools of engineering to design the system or components of your project (PO5,PSO1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
5.	Identification of impact in societal and environmental contexts of your project developed in interdisciplinary domain (PO7,PSO2)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
6.	Are you responsible, diligent, and ethical on your project(core and interdisciplinary) development period with your team members? (PO8, PSO2)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
7.	Have you functioned effectively as a member in a team with your project members? (PO9)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
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8.	Have you assessed any societal, health, safety, legal and cultural issues in your project? (PO6)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
9.	Have you presented your project confidently and prepared effective reports? (PO10)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
10.	Have you completed the project within a time frame and estimated budget? (PO11)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
11.	Is your project leads you to recognize the need for lifelong learning in the context of technological change? (PO12)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
12.	Have you conducted the experiments/ analysis to provide valid conclusion? (PO4,Pso1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1

Signature of the Student:

Date:



AMC ENGINEERING COLLEGE
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

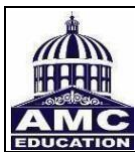
PROJECTWORK SELF EVALUATION FORM

Name	PALLAVI S	USN	1AM22EC084			
Program	Bachelor of Engineering in Electronics and Communication					
S.No	To what extent did each of the following you contribute to your project	To a very great extent	To a great extent	To a moderate extent	To some extent	Not at all
1.	Have you applied the knowledge of mathematics, science and engineering concepts(PO1,PSO1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
2.	Have you analyzed the system of hardware, software and its interaction (PO2,PSO1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
3.	Design the systems, components or processes to meet needs(PO3,PSO1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
4.	Have you used the modern tools of engineering to design the system or components of your project (PO5,PSO1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
5.	Identification of impact in societal and environmental contexts of your project developed in interdisciplinary domain (PO7,PSO2)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
6.	Are you responsible, diligent, and ethical on your project (core and interdisciplinary) development period with your team members? (PO8, PSO2)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
7.	Have you functioned effectively as a member in a team with your project members? (PO9)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
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8.	Have you assessed any societal, health, safety, legal and cultural issues in your project? (PO6)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
9.	Have you presented your project confidently and prepared effective reports? (PO10)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
10.	Have you completed the project within a time frame and estimated budget? (PO11)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
11.	Is your project leads you to recognize the need for lifelong learning in the context of technological change? (PO12)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
12.	Have you conducted the experiments/ analysis to provide valid conclusion? (PO4,Pso1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1

Signature of the Student:

Date:



AMC ENGINEERING COLLEGE
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

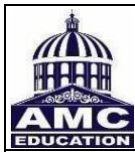
PROJECTWORK SELF EVALUATION FORM

Name	PRADEEP S N	USN	1AM22EC086			
Program	Bachelor of Engineering in Electronics and Communication					
S.No	To what extent did each of the following you contribute to your project	To a very great extent	To a great extent	To a moderate extent	To some extent	Not at all
1.	Have you applied the knowledge of mathematics, science and engineering concepts(PO1,Pso1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
2.	Have you analyzed the system of hardware, software and its interaction (PO2,PSO1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
3.	Design the systems, components or processes to meet needs(PO3,PSO1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
4.	Have you used the modern tools of engineering to design the system or components of your project (PO5,PSO1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
5.	Identification of impact in societal and environmental contexts of your project developed in interdisciplinary domain (PO7,PSO2)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
6.	Are you responsible, diligent, and ethical on your project(core and interdisciplinary) development period with your team members? (PO8, PSO2)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
7.	Have you functioned effectively as a member in a team with your project members? (PO9)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
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8.	Have you assessed any societal, health, safety, legal and cultural issues in your project? (PO6)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
9.	Have you presented your project confidently and prepared effective reports? (PO10)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
10.	Have you completed the project within a time frame and estimated budget? (PO11)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
11.	Is your project leads you to recognize the need for lifelong learning in the context of technological change? (PO12)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
12.	Have you conducted the experiments/ analysis to provide valid conclusion? (PO4,Pso1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1

Signature of the Student:

Date:



AMC ENGINEERING COLLEGE
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

PROJECTWORK SELF EVALUATION FORM

Name	PRATHIBHA H	USN	1AM22EC089			
Program	Bachelor of Engineering in Electronics and Communication					
S.No	To what extent did each of the following you contribute to your project	To a very great extent	To a great extent	To a moderate extent	To some extent	Not at all
1.	Have you applied the knowledge of mathematics, science and engineering concepts(PO1,Pso1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
2.	Have you analyzed the system of hardware, software and its interaction (PO2,PSO1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
3.	Design the systems, components or processes to meet needs(PO3,PSO1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
4.	Have you used the modern tools of engineering to design the system or components of your project (PO5,PSO1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
5.	Identification of impact in societal and environmental contexts of your project developed in interdisciplinary domain (PO7,PSO2)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
6.	Are you responsible, diligent, and ethical on your project(core and interdisciplinary) development period with your team members? (PO8, PSO2)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
7.	Have you functioned effectively as a member in a team with your project members? (PO9)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
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8.	Have you assessed any societal, health, safety, legal and cultural issues in your project? (PO6)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
9.	Have you presented your project confidently and prepared effective reports? (PO10)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
10.	Have you completed the project within a time frame and estimated budget? (PO11)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
11.	Is your project leads you to recognize the need for lifelong learning in the context of technological change? (PO12)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1
12.	Have you conducted the experiments/ analysis to provide valid conclusion? (PO4,Pso1)	☐ 5	☐ 4	☐ 3	☐ 2	☐ 1

Signature of the Student:

Date: